

Group 3200_21

Minutes of Client Meeting 1

30/07/2026

Present: Christo, Lithasa, Dhava, Oliver, Adrian, Dylan

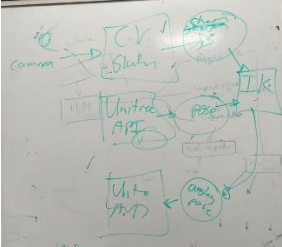
Apologies: Murphy, Terry

Absentees:

Meeting Start: 1.30pm

Meeting End: 2.30pm

Time:	Item #	Topic:	Details:	Action:
1330	1	Introductions	Everyone introduced themselves	
1335	2	Initial Questions	Client answered some initial questions: - We are using the Python version of the Unitree SDK to 'talk' to the robot. The SDK language will not affect anything. - The product will be a program to 'run the robot' - o The robot will mimic and copy people's actions. - o The robot will also respond to action commands informing it to complete a task (i.e. wave hand in a circle -> robot turns around) - Clientele / target audience would be real people in a physio / health science environment who will mimic the program the robot runs. - Our first milestone is to develop a working understanding of how everything is put together and to research the different topics explained to us - Before testing the program on the robot, testing will be done on a simulator first. - We are recommended to use Ubuntu 22 instead of 20 for development. - We are also recommended to set up a docker with Ubuntu 22 inside it for development on each of our machines.	
1355	3	Topics to learn and	To Learn: - G1 SDK and Development documentation - General Motion Retargeting (GMR) (Github) - o Paired with Xsens motions capture suit - Generate Skeleton from Camera input - Kinematics (Concepts) - o Transformation metrics - o Forward Kinematic - o Inverse kinematics - Posture / action recognition (Pinocchio library) - Use of Git/Github and Docker - Ubuntu 22	- ALL members are to research these topics. - ALL members are to implement a Ubuntu 22 Docker container on your device ready for development

		Explanation of Concepts involved	 Data in: - Camera stream => 'Translator' => 'Skeleton' (connected points in 3D space depicting a human) => IK - Unitree API => Current robot pose => IK - Motion capture suit => 'GMR' => 'Skeleton' => IK Data out: - Inverse Kinetics (IK) => analog pose => Unitree API	
1420	4	Personal questions and clarification	Present members clarified personal questions and interest questions	
1430	5	Meeting end		